

Inhibitor Tolerance: the Data

Vanacloig-Pedros 2022 and Hillenmeyer 2008 axis by axis, their replication and ceilings, the response structure they share, and the molecular encoders

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Abstract

Vanacloig-Pedros 2022 is the one genome-wide screen in the served graph that doses the industrial inhibitor panel, furfural, 5-hydroxymethylfurfural, the phenolic aldehydes and acids, the ionic liquids, ethanol and isobutanol, at a matched IC₃₀, and it is the only dataset in which isobutanol appears as a stress. Hillenmeyer 2008 is the only other genome-wide environment-response compendium of comparable size. Every number here is read from the served records of the two datasets through their loaders. The two agree on genes (3,550 of Vanacloig’s 3,598 queried genes are in the homozygous Hillenmeyer arm) and disagree on almost everything else: Vanacloig strains are quadruple mutants on a drug-sensitized host, anaerobic, in a defined hydrolysate mimic at pH 5, dosed to a target inhibition; Hillenmeyer strains are single deletions in diploids, aerobic, in YPD, dosed by mass. Only two compounds are shared per arm.

Replication sets the ceiling. Vanacloig’s three batch replicates agree at a median Spearman of 0.33 over genes and its bottom-5% hit sets at a Jaccard of 0.24; the served-standard-error reliability index tracks that agreement at rank 0.87 and gives a median ceiling of 0.84 on the correlation any model can reach with the noise-free response, 0.84 for isobutanol and 0.80 for furfural, and no ceiling at all for six compounds including 5-hydroxymethylfurfural. For Hillenmeyer the served standard error overstates reliability (index 0.92 against a raw replicate agreement of 0.39), so its raw replicate agreement is the number to use. The response structure the datasets share is real and small: a per-gene mean-response correlation of 0.118 and a gene-gene similarity agreement eleven null standard deviations from zero for the homozygous arm, a third of that for the heterozygous arm, and exactly one compound, methyl methanesulfonate, whose per-gene profile transfers. Benomyl, the acetate salt against the free acid, and ferulic acid at two doses do not. Molecular encoders embed every compound of the three datasets, and whether chemical similarity predicts response similarity where exact matches do not exist is measured for each encoder in Section 6.

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1 The question x

A strain-design model for industrial hosts has to predict how a genotype responds to the inhibitors a real feedstock carries. Lignocellulosic hydrolysate brings furans (furfural, 5-hydroxymethylfurfural), weak acids (acetic, levulinic, formic), phenolics from lignin (vanillin, syringaldehyde, ferulic and p-coumaric acid), and the ionic liquids left by pretreatment; a producing strain also has to tolerate its own product, which for the isobutanol work is isobutanol. Vanacloig-Pedros et al. 2022 (10.1093/femsyr/foac036) screened a barcoded deletion library against exactly that panel, and it is the only genome-wide dataset in the served graph that does. The question for a model is what other data can help it, and the candidate is the only other genome-wide environment-response compendium of comparable size, Hillenmeyer et al. 2008 (10.1126/science.1150021), in its homozygous (HOM) and heterozygous (HET) arms.

Every script named below lives in `experiments/031-env-chemgen-inhibitor-tolerance/scripts/`. Three things have to be known before a model is trained on the pair. First, the axes on which the datasets differ, because an environment encoding has to carry every axis that changes between them and a cell encoding has to carry every axis that changes inside the strain. Second, how well each dataset agrees with itself, because replicate agreement is the ceiling on any prediction and it is not the same for every compound. Third, how much response structure the two datasets share, because a partner that measures different biology adds parameters and no signal.

Vocabulary. A **condition** is one dosed compound, or one physical factor or temperature when no compound is dosed; Hillenmeyer’s several doses and generation counts of one compound are one condition here. The **response** is the served number for a strain in a condition, oriented throughout so that negative is a fitness defect. A **hit** is a gene in the bottom 5% of responses in a condition. The **reliability index** of a condition is $1 - \overline{SE^2}/\text{Var}(\text{response})$ over its genes, the share of the across-gene variance that the served replicate noise cannot produce. **Replicate agreement** is the Spearman correlation over genes between two raw replicates of a condition, and **hit agreement** the Jaccard index of their hit sets. The **cross Spearman** of a Vanacloig condition and a Hillenmeyer condition is the Spearman correlation of their responses over the genes both screened. An **exact match** is a compound both datasets dose under the same InChIKey; **chemical similarity** is Tanimoto similarity of fingerprints or cosine similarity of a learned embedding between two compounds that are not the same molecule.

Measured. Every axis of the three datasets from their served records (Section 2); replication, served error and per-condition ceilings (Section 3); the shared response structure and the transfer of every exact-match compound (Section 4); the coverage of every molecular encoder over the three compound sets and the relation between chemical similarity and response similarity (Sections 5 and 6).

Not measured. No model has been trained. The linear and nearest-neighbor baselines that turn these ceilings into scores are Section 7.

2 The three datasets, axis by axis x

Every record of the three stores was flattened to one row through its loader (`flatten\_records.py`), so each value below is the served pydantic record, not a raw-matrix value. Table 1 holds the axes a cell and environment representation has to encode.

The cell axis. A Vanacloig genotype is never a single deletion. The library is the drug-sensitized *3DeltaAlpha* collection, so every strain carries the queried barcoded deletion on top of *pdr1*, *pdr3* and *snq2* deletions, which the loader stores as three constant background perturbations: four genes per genotype, haploid, on S288C. Hillenmeyer HOM is one KanMX deletion in a homozygous diploid; HET is one copy of a gene removed from a diploid, a dosage halving that reaches the essential genes HOM cannot. A model that reads the genotype as a gene set therefore sees four genes per Vanacloig strain and one per Hillenmeyer strain, and the three background genes are absent from Vanacloig’s queried set. The cell representation differs on background, ploidy and perturbation class at once, before any environment is considered.

The environment axis. Two axes are constant within each dataset and opposite between them. Vanacloig is anaerobic in SynBase, a defined synthetic hydrolysate mimic at pH 5.0 set with HCl; Hillenmeyer is aerobic in YPD on 86% of HOM records and 99% of HET records, with SC dropouts, SD and YP glycerol for the rest. Neither oxygen nor base medium varies inside its own dataset, so a joint model cannot learn their effect from these two datasets; it can only learn a dataset offset. Temperature is 30 C on every Vanacloig record and a typed provenance gap on 98% of Hillenmeyer records. Duration is in generations on both sides on different scales, 6.5 against mostly 20. Dose is the third asymmetry: Vanacloig records an IC30 basis with no molar value for 39 of its 41 compounds, Hillenmeyer a numeric dose on 98% of records, so a dose feature is a different kind of number in the two sources. Every Vanacloig

Table 1: The cell, environment and readout axes of the three datasets, from the served records. Each cell lists the distinct values with the number of records carrying them. Vanacloig 2022 carries one value on every environment axis; Hillenmeyer varies medium, temperature and duration inside the dataset.

axis	Vanacloig 2022	Hillenmeyer HOM	Hillenmeyer HET
records	143,218	1,088,620	2,698,797
queried genes	3,598	4,675	5,825
genes per genotype	4 (143,218)	1 (1,088,620)	1 (2,698,797)
ploidy	haploid (143,218)	diploid (1,088,620)	diploid (2,698,797)
compounds	41	116	302
conditions	41	123	307
environments	41	257	474
medium base	SYNH3_MINUS (143,218)	YPD (935,890); SC (90,587); SD (53,524)	YPD (2,664,192); SD (23,133); YP (11,472)
oxygen	anaerobic (143,218)	aerobic (1,088,620)	aerobic (2,698,797)
temperature (C)	30.0 (143,218)	None (1,062,895); 25.0 (8,619); 23.0 (8,619)	None (2,670,237); 23.0 (11,472); 37.0 (11,411)
duration (generations)	6.5 (143,218)	20.0 (654,439); 5.0 (295,904); 10.0 (69,370)	20.0 (2,509,318); 5.0 (91,855); 15.0 (86,070)
dose basis	IC30 (136,237); fixed (6,981)	(1,067,628); , (12,815); fixed (8,177)	(2,545,140); , (147,953); fixed (5,704)
statistic	log2_ratio (143,218)	z_score (1,088,620)	log2_ratio (2,698,797)
replicates per record	3 (143,218)	1 (711,216); 2 (264,441); 3 (78,091)	1 (2,043,883); 2 (467,120); 3 (96,723)

record carries exactly one dosed compound; 18% of HOM records carry none (media swaps, temperature, radiation, pH) and 1% carry two.

The readout axis. All three are pooled competitive-growth barcode assays, but the statistic differs: Vanacloig stores a log2 ratio of treated to control CPM, averaged over three batch replicates with a sample SD on every record; HOM stores a z-score against a matched control set and HET a log2 ratio, with a served SD on 35% and 24% of records because 65% and 76% of records come from a single array. The sign is opposite to Vanacloig’s (positive is a defect in Hillenmeyer). The label must be per dataset, or reduced to a sign or a rank, before anything is pooled.

Overlap. Genes overlap almost completely and compounds almost not at all (Table 2). Sodium acetate (Vanacloig) and acetic acid (Hillenmeyer) are different compound entities and do not join; the same holds for sodium butyrate against any butyric acid record. Benomyl is dosed at 10 $\mu\text{g}/\text{mL}$ on both sides. Hillenmeyer’s ferulic acid is 3.95 μM in HET only, against Vanacloig’s IC30.

Table 2: Queried-gene and compound overlaps. Queried genes exclude Vanacloig’s three constant background genes. Compounds are keyed by InChIKey, so a salt and its free acid are different entries.

set	pair	n_a	n_b	shared	Jaccard
queried genes	Vanacloig 2022 and Hillenmeyer 2008 HOM	3,598	4,675	3,550	0.75
queried genes	Vanacloig 2022 and Hillenmeyer 2008 HET	3,598	5,825	3,577	0.61
queried genes	Hillenmeyer 2008 HOM and Hillenmeyer 2008 HET	4,675	5,825	4,648	0.79
queried genes	all three			3,533	
compounds (InChIKey)	Vanacloig 2022 and Hillenmeyer 2008 HOM	41	114	2	0.01
compounds (InChIKey)	Vanacloig 2022 and Hillenmeyer 2008 HET	41	290	2	0.01
compounds (InChIKey)	Hillenmeyer 2008 HOM and Hillenmeyer 2008 HET	114	290	99	0.32
compounds (InChIKey)	all three			1	

3 Replication, error, and the ceiling on prediction $\times$

Two views of noise are available for every condition, and where they can be joined they check each other (`replicate\_noise\_and\_ceilings.py`).

The served view. The served record carries the replicate count and, where the source released it, a sample SD from which the SE is derived. Per condition the reliability index $\text{rel} = 1 - \overline{\text{SE}}^2 / \text{Var}(\text{response})$ over genes is the share of the across-gene spread that replicate noise cannot produce. Under the usual measurement model, $\text{response} = \text{truth} + \text{noise}$ with the noise independent of the truth, $\sqrt{\text{rel}}$ is the ceiling on the Pearson correlation between a measurement and the noise-free truth, and rel itself is the ceiling on the correlation between two independent measurements of the same condition, which is what a held-out replicate, or a second dataset, can reach.

The empirical view. The raw matrices the loaders consumed hold the replicates the served SE summarizes: Vanacloig’s three batches, each a log2 CPM ratio against its own batch’s control columns (the loader’s quantity), and Hillenmeyer’s replicate arrays of one condition label. Pairwise Spearman between replicates over genes, and the Jaccard of their bottom-5% hit sets, are the model-free reliability of a condition and of its hits. Raw labels are mapped to served compound names with the loaders’ own resolver, so the two views join.

Table 3: Replication and error per dataset. The reliability index and the ceiling on r come from the served SE; the replicate Spearman and hit Jaccard from the raw replicates. The last row is the rank correlation between the two views over the conditions that have both.

	Vanacloig 2022	Hillenmeyer HOM	Hillenmeyer HET
records	143,218	1,088,620	2,698,797
records with one replicate	0%	65%	76%
records with a served SE	100%	35%	24%
SE median	0.138	0.344	0.096
response SD	0.699	3.529	0.439
conditions	41	140	308
reliability index, median	0.703	0.919	0.845
reliability index, 25th pct	0.370	0.804	0.668
ceiling on r vs truth, median	0.838	0.959	0.919
conditions with no signal	3	0	2
raw conditions with replicates	41	78	108
replicate Spearman, median	0.333	0.386	0.255
bottom-5% hit Jaccard, median	0.237	0.203	0.146
index vs replicate Spearman (rank)	0.87	0.40	0.43

What the two views say. Vanacloig’s three batches agree at a median Spearman of 0.33 and its hit sets at a Jaccard of 0.24 (Table 3, Figure 1b,c). Its served index tracks the raw agreement at a rank correlation of 0.87 (Figure 2d), so the served SE is an honest per-compound noise estimate for this dataset. For Hillenmeyer it is not: the served index has a median of 0.92 for HOM and 0.84 for HET while the raw replicate arrays agree at 0.39 and 0.26, and the two views correlate at only 0.40 and 0.43. Plausible reason, unverified: the served SD exists only for the third of records with more than one array and describes array-to-array scatter within one control set, while the raw replicates of a label span control sets and scanners. For Hillenmeyer the raw replicate agreement is the ceiling to quote.

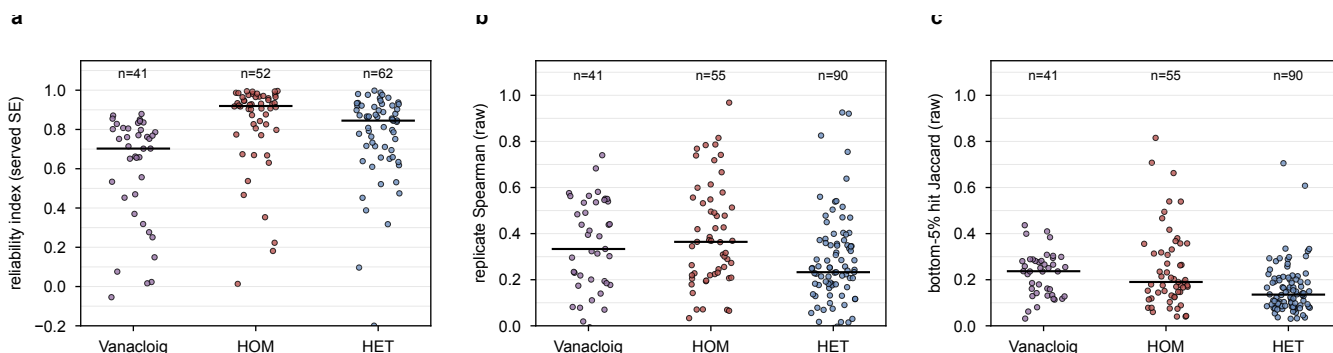


Figure 1: Noise per condition in the three datasets. (a) The served-SE reliability index, one point per condition, bar at the median. (b) Pairwise Spearman between raw replicates of a condition over genes. (c) Jaccard of the bottom-5% hit sets between raw replicates. Vanacloig has 41 conditions in every panel; Hillenmeyer’s counts differ between panels because the served SE exists only where a source SD was released and raw replicates only where a label was run on two or more arrays.

The ceiling per Vanacloig compound. Figure 3 and Table 4 give every Vanacloig compound its ceiling. Methyl methanesulfonate, 2,2’-bipyridine, crystal violet, ferulic acid and methylglyoxal sit above 0.92. Isobutanol is at 0.89, furfural at 0.88, ethanol at 0.91. Six compounds have an index below 0.1, and three of those (sodium glyoxylate, sodium butyrate, 5-hydroxymethylfurfural) are at or below zero: at the doses used, their replicate scatter is as large as the spread between genes, and there is nothing for a model to predict. 5-hydroxymethylfurfural is the one loss that matters for the hydrolysate question.

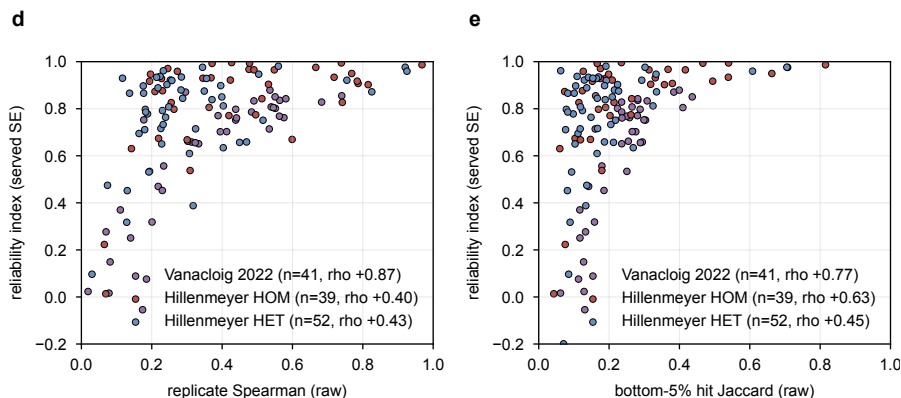


Figure 2: The served index against the raw agreement, one point per condition. (d) Against replicate Spearman. (e) Against hit Jaccard. The legend carries the rank correlation per dataset. Vanacloig’s index is a faithful summary of its replicate agreement; Hillenmeyer’s overstates it.

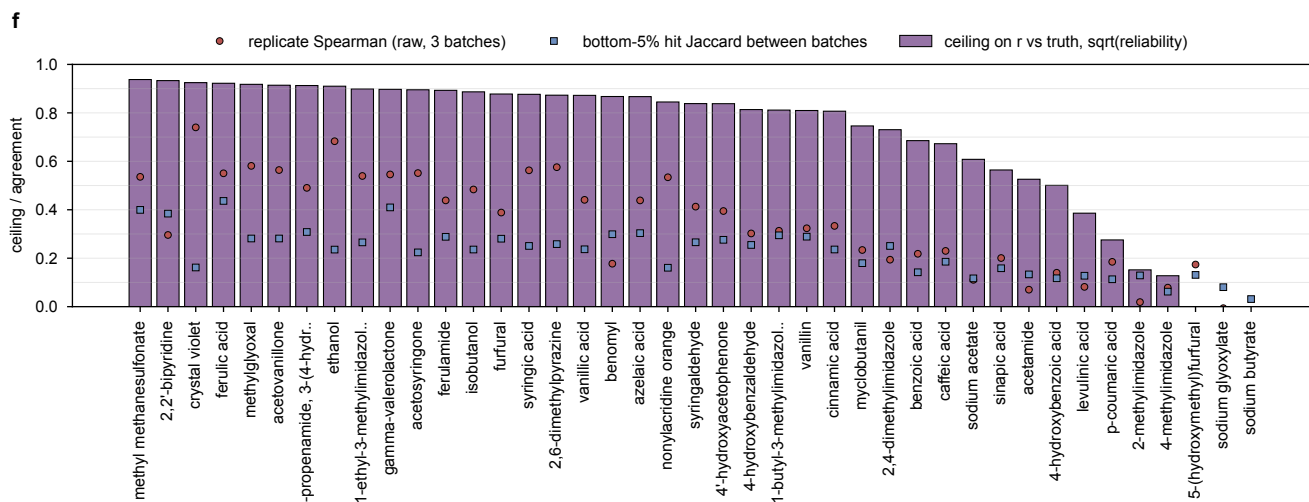


Figure 3: (f) Every Vanacloig compound, ordered by its ceiling on the correlation with the noise-free response (bars, $\sqrt{\text{rel}}$), with the raw three-batch replicate Spearman (circles) and the between-batch hit Jaccard (squares). A compound with no bar has an index at or below zero.

Table 4: Every Vanacloig compound: served reliability index, ceiling on r versus the truth, raw replicate Spearman over three batches, between-batch hit Jaccard, and the best-correlated Hillenmeyer HOM condition with its cross Spearman.

compound	reliability	ceiling r	replicate ρ	hit Jaccard	best HOM match	ρ
methyl methanesulfonate	0.88	0.94	0.54	0.40	4-nitroquinoline 1-oxide	0.15
2,2'-bipyridine	0.87	0.93	0.30	0.38	CID 622648	0.11
crystal violet	0.86	0.92	0.74	0.16	medium=SC - thiamine h..	0.13
ferulic acid	0.85	0.92	0.55	0.44	cis-oxalato-trans-l-1,..	0.08
methylglyoxal	0.84	0.92	0.58	0.28	medium=SC, partial bio..	0.10
acetovanillone	0.84	0.91	0.56	0.28	methyl methanesulfonate	0.15
2-propenamide, 3-(4-hy..	0.83	0.91	0.49	0.31	nocodazole	0.10
ethanol	0.83	0.91	0.68	0.24	basifungin	0.15
1-ethyl-3-methylimidaz..	0.81	0.90	0.54	0.27	1-methyl-4-phenylpyrid..	0.12
gamma-valerolactone	0.80	0.90	0.55	0.41	basifungin	0.16
acetosyringone	0.80	0.90	0.55	0.22	methyl methanesulfonate	0.13
ferulamide	0.80	0.89	0.44	0.29	mechlorethamine	0.15
isobutanol	0.79	0.89	0.48	0.24	basifungin	0.13
furfural	0.77	0.88	0.39	0.28	basifungin	0.13
syringic acid	0.77	0.88	0.56	0.25	nystatin [usp:inn:ban:..	0.11
2,6-dimethylpyrazine	0.76	0.87	0.58	0.26	bleomycin	0.09
vanillic acid	0.76	0.87	0.44	0.24	calyculin a	0.13
benomyl	0.75	0.87	0.18	0.30	potassium metabisulfite	0.10
azelaic acid	0.75	0.87	0.44	0.30	parthenolide	0.11
nonylacridine orange	0.71	0.84	0.53	0.16	medium=SC - thiamine h..	0.17
syringaldehyde	0.70	0.84	0.41	0.27	T=37	0.06
4'-hydroxyacetophenone	0.70	0.84	0.39	0.28	nocodazole	0.08
4-hydroxybenzaldehyde	0.66	0.81	0.30	0.25	nocodazole	0.09
1-butyl-3-methylimidaz..	0.66	0.81	0.31	0.29	medium=SC - L-tryptophan	0.11
vanillin	0.66	0.81	0.32	0.29	methyl methanesulfonate	0.08
cinnamic acid	0.65	0.81	0.33	0.24	platinum(ii),1-cyclobu..	0.10
myclobutanil	0.56	0.75	0.23	0.18	basifungin	0.22
2,4-dimethylimidazole	0.53	0.73	0.19	0.25	potassium metabisulfite	0.08
benzoic acid	0.47	0.69	0.22	0.14	nocodazole	0.10
caffeic acid	0.45	0.67	0.23	0.19	nystatin [usp:inn:ban:..	0.15
sodium acetate	0.37	0.61	0.11	0.12	sodium fluoride	0.09
sinapic acid	0.32	0.56	0.20	0.16	T=37	0.10
acetamide	0.28	0.53	0.07	0.13	busulfan	0.08
4-hydroxybenzoic acid	0.25	0.50	0.14	0.12	camptothecin	0.08
levulinic acid	0.15	0.39	0.08	0.13	basifungin	0.08
p-coumaric acid	0.08	0.28	0.18	0.11	curcumin	0.13
2-methylimidazole	0.02	0.15	0.02	0.13	streptozocin	0.08
4-methylimidazole	0.02	0.13	0.08	0.06	methyl methanesulfonate	0.08
5-(hydroxymethyl)furfu..	-0.05	0.00	0.17	0.13	rotenone	0.07
sodium butyrate	-0.47	0.00	-0.07	0.03	medium=SC - L-threonine	0.16
sodium glyoxylate	-1.02	0.00	-0.01	0.08	basifungin	0.17

4 What Hillenmeyer shares with Vanacloig x

Both matrices are oriented so that negative is a defect, Hillenmeyer’s doses and generation counts are averaged within a compound, and every Vanacloig condition is correlated with every partner condition over the shared genes (`cross\_dataset\_similarity.py`). Three statistics ask three different questions. The cross Spearman of two conditions asks whether the same genes are sensitive to both. The per-gene mean response across all conditions asks whether the datasets agree on which genes are generally sick. The Mantel statistic asks whether they agree on which genes move together: the two gene-by-gene correlation matrices, built from each dataset’s condition profiles, are compared entry by entry against a null that shuffles gene identity.

Table 5: Shared response structure between Vanacloig and each Hillenmeyer arm. The cross Spearman columns summarize the full condition-by-condition matrix; the Mantel column is the Spearman between the two gene-gene profile-correlation matrices on 1,500 sampled genes, with the mean and SD of 20 gene-permutation nulls.

arm	conditions	shared genes	cross ρ median	95th pct	max	mean-response ρ	Mantel ρ (null)
HOM	140	3,550	0.010	0.075	0.222	0.118	0.068 (-0.003 $\pm$ 0.006)
HET	308	3,577	0.003	0.052	0.183	0.036	0.024 (-0.001 $\pm$ 0.007)

The shared structure is real and small (Table 5). For HOM the Mantel statistic is eleven null standard deviations from zero and the per-gene mean response correlates at 0.118, while the median cross Spearman of a condition pair is 0.010 and the largest is 0.222. HOM carries about three times the shared structure that HET does on every statistic, which is the expected direction: HOM is the same perturbation class as Vanacloig, HET is a dosage halving.

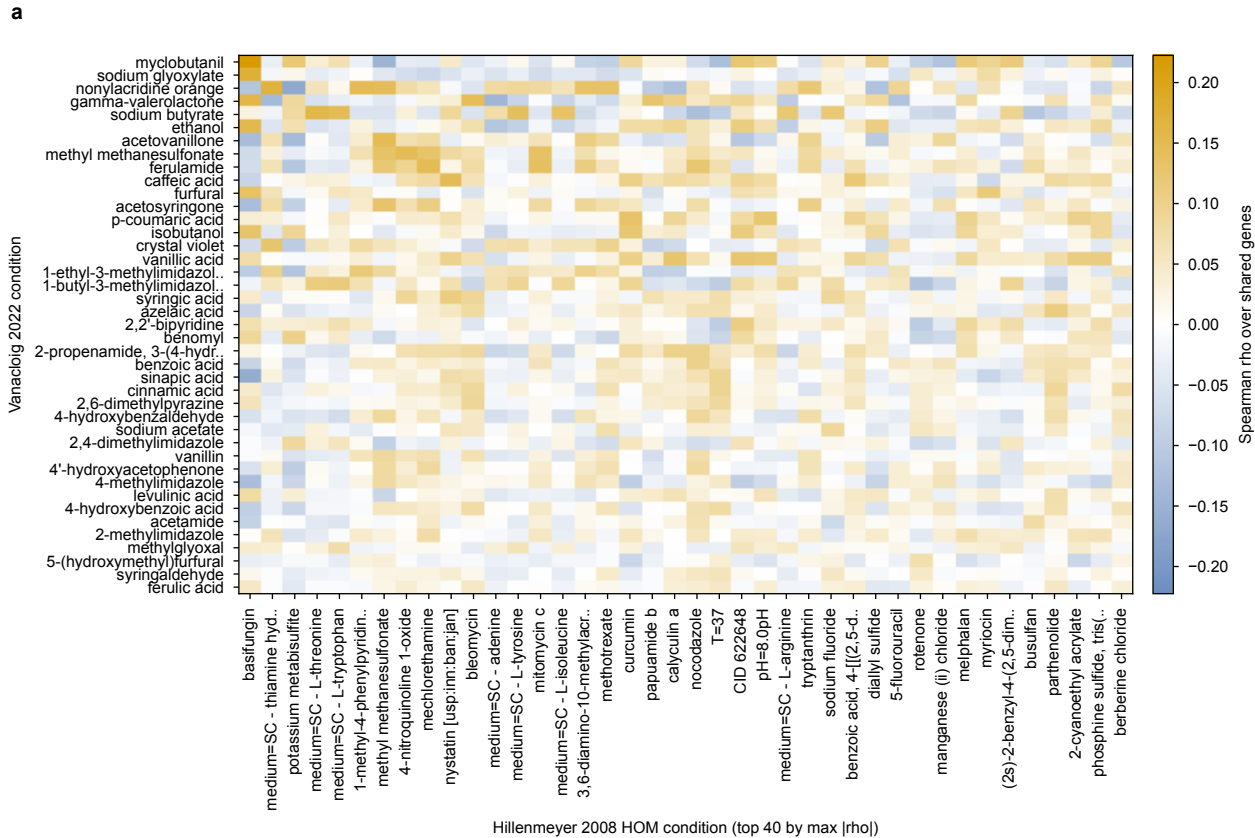


Figure 4: (a) Cross Spearman between each Vanacloig condition (rows) and the 40 HOM conditions with the largest absolute correlation to any row (columns), over the 3,550 shared genes. The solvent-like inhibitors (ethanol, isobutanol, furfural, gamma-valerolactone) and myclobutanil share their best match, basifungin; the phenolic aldehydes and ketones match methyl methanesulfonate and mechlorothamine.

The exact-match compounds. Two compounds are shared with each arm, and Table 6 gives each pair the same three tests. The cross Spearman is the correlation of the two per-gene profiles. The rank places the true match among all partner conditions: if the same compound in the other dataset is the best-correlated condition, the rank is 1. The hit overlap is the number of genes in the bottom 5% of both profiles, against the 8.6 expected by chance, with a one-sided

Fisher test.

Table 6: Transfer of every compound both datasets dose, plus the sodium acetate versus acetic acid near-pair, which the graph does not join and which serves as the negative control. Rank is the position of the true match among all partner conditions by cross Spearman; hits is the overlap of bottom-5% genes with the chance expectation in parentheses.

compound (V / H)	arm	dose V / H	genes	ρ	rank	hits (exp.)	Fisher p
benomyl	HOM	10.0 $\mu\text{g/mL}$ / 10.0 $\mu\text{g/mL}$	3,428	-0.007	88 of 140	16 (8.6)	1.1e-02
methyl methanesulfonate	HOM	fixed / 0.002 % v/v; 0.004 % v/v	3,426	0.132	3 of 140	39 (8.6)	8.8e-17
sodium acetate / acetic acid	HOM	IC30 / IC30 vs acid	3,256	-0.027	121 of 140	4 (8.2)	9.7e-01
benomyl	HET	10.0 $\mu\text{g/mL}$ / 13.8 μM ; 2.4 μM ; 27.6 μM ; 3.4 μM ; 6.9 μM	3,436	0.024	30 of 308	5 (8.6)	9.4e-01
ferulic acid	HET	IC30 / 3.95 μM	3,416	0.005	110 of 308	8 (8.6)	6.3e-01

Figure 5 shows what the table summarizes. Each panel is one shared compound; each point is one gene, placed at its Vanacloig response (x) and its HOM response (y), both oriented so that lower is sicker. A gene that the compound hurts in both screens falls toward the lower left, and the red points are the genes in the bottom 5% of both. Methyl methanesulfonate (panel d) is what transfer looks like: a cloud that leans toward the lower left, 39 genes hit in both against 8.6 expected, and the true match third of 140 HOM conditions, behind two other DNA-damaging agents. Benomyl (panel c) is what a hit-level-only transfer looks like: the cloud is round, the correlation is zero, and the 16 doubly hit genes are the only structure. Sodium acetate against acetic acid (panel e) is what no transfer looks like: 4 shared hits, fewer than chance.

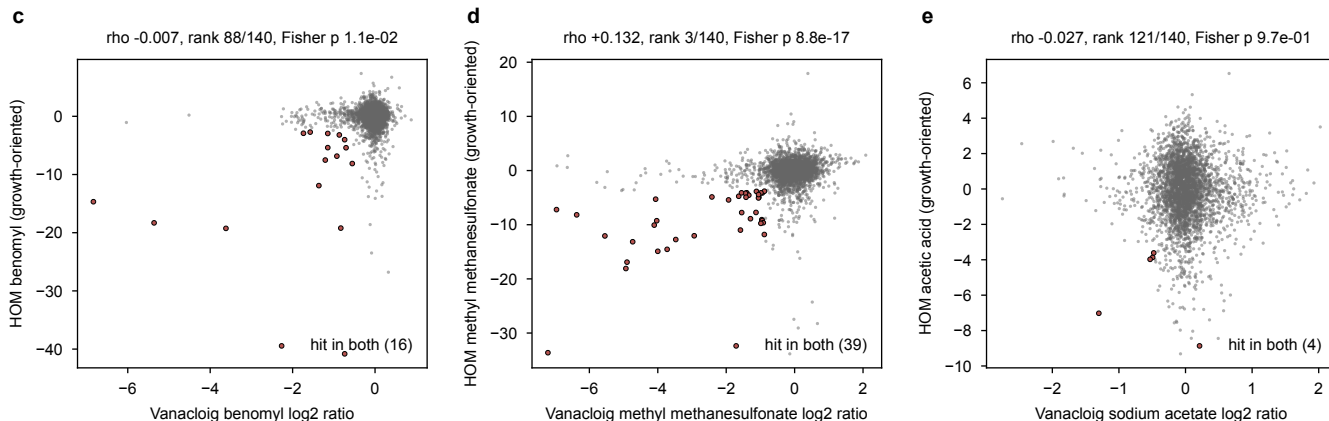


Figure 5: Per-gene responses for the three HOM pairs, one point per shared gene, bottom-5% hits in both datasets in red. (c) Benomyl at 10 $\mu\text{g/mL}$ in both. (d) Methyl methanesulfonate, fixed dose in Vanacloig against 0.002 and 0.004% in HOM. (e) Sodium acetate at IC30 in anaerobic SynBase against acetic acid in YPD.

The best matches group by chemistry without a shared compound. The full ranking (`results/top\_matches\_hom.csv`) puts basifungin first for ethanol, isobutanol, furfural, gamma-valerolactone, sodium glyoxylate and myclobutanil, with myriocin second for furfural and glyoxylate; acetovanillone, acetosyringone and the phenolic amides match methyl methanesulfonate and mechlorethamine first. Hypothesis (untested): the solvent-like inhibitors share a membrane or sphingolipid stress profile and the aldehyde and ketone phenolics share a DNA-damage profile. Section 6 asks the general form of this question with molecular encoders.

5 Molecular encoders and their coverage x

A compound enters a model as a vector, and the choice of encoder is a modeling decision that should be made on measured coverage and measured behavior rather than on a reading list. Twelve small-molecule encoders now live in `torchcell/molecule/`: a `MoleculeEncoder` turns a list of SMILES into one fixed-width float32 vector per molecule, raises on a SMILES it cannot parse rather than returning zeros, and is deterministic across calls. Four are RDKit fingerprints and descriptors with no learned weights (count and bit ECFP4 at 2,048, feature-class FCFP4, MACCS keys, and the 217 RDKit 2D descriptors); one is Mol2Vec, a skip-gram over Morgan substructure identifiers; four are SMILES transformers (the two ChemBERTa-2 heads, MolFormer-XL, a ZINC RoBERTa); one is Uni-Mol, which embeds an ETKDG conformer; and one is MolE, the contrastively pretrained graph network built for antimicrobial activity. Weights that are not on the Hugging Face hub (Mol2Vec, MolE) sit under `\$DATA\_ROOT` with a pydantic

manifest recording the source URL, the retrieval command and the sha256, re-hashed on every load. The registry and its tests are described in the dendron note `torchcell.molecule.encoders`.

Why these twelve. Three principles chose the set. The benchmark of 25 pretrained embeddings on 25 datasets found almost every one statistically indistinguishable from count ECFP [▲](#) [external: arXiv 2508.06199, September 2026 revision], so fingerprints are the mandatory baseline and every learned encoder has to earn its place against them. Everything installs from RDKit, pip or the Hugging Face hub, with one vendored exception. And the four representation families are each present, so the transfer test can say which family carries chemistry-to-response signal rather than which checkpoint won.

- **Fingerprints and descriptors** (count and bit ECFP4, FCFP4, MACCS, the 217 RDKit descriptors): no training corpus, hashed or curated substructures and physicochemical properties [▲](#) [external: Rogers and Hahn 2010, 10.1021/ci100050t; Durant et al. 2002, 10.1021/ci010132r]. FCFP4 tests whether pharmacophore feature classes beat atom identity; the descriptors are what a tolerance biologist reasons with (logP, molecular weight, polar surface area).
- **Mol2Vec**: skip-gram over Morgan identifiers on 19.9 million ZINC and ChEMBL compounds, 300 dimensions, still the metabolomics workhorse [▲](#) [external: Jaeger, Fulle and Turk 2018, 10.1021/acs.jcim.7b00616].
- **SMILES transformers**: ChemBERTa-2 in its masked-language and multitask-regression heads on 77 million PubChem SMILES [▲](#) [external: Ahmad et al. 2022, arXiv 2209.01712], MolFormer-XL on 1.1 billion PubChem and ZINC SMILES with linear attention, of which the public checkpoint carries the 10 percent subset [▲](#) [external: Ross et al. 2022, 10.1038/s42256-022-00580-7], and a community ZINC RoBERTa at 480 million SMILES added for availability. The two ChemBERTa heads share a backbone, which separates SMILES syntax from property supervision.
- **3D**: Uni-Mol, pretrained on 209 million conformers by atom masking, coordinate denoising and pair-distance prediction [▲](#) [external: Zhou et al. 2023, ICLR]; the one encoder that can say whether conformation matters for the small solvents and the phenolics.
- **Graph**: MolE, a GIN pretrained with a Barlow Twins objective on 100,000 PubChem structures and validated experimentally for antimicrobial discovery [▲](#) [external: Olayo-Alarcon et al. 2025, 10.1038/s41467-025-58804-4]; whole-cell growth inhibition is the published task closest to inhibitor tolerance, and its model code is vendored under `third\_party/mole/`.

Left out. Chemprop’s D-MPNN is trained end to end and belongs to the model phase as an architecture, not to an embedding comparison. CLAMP, the strongest frozen model in the benchmark, aligns fingerprints to mammalian and enzymatic assay text and ships only as a research repository; it is the first candidate for a second pass if the fingerprint arm wins. MolCLR, GraphMVP, 3D Infomax, Mole-BERT and KPGT are GIN checkpoints of MolE’s family in research repositories; GEM is PaddlePaddle only; CLOOME, MoCoP and InfoAlign encode human Cell Painting or expression phenotypes whose transfer to yeast growth is untested; MoleculeSTM, MolBERT, R-MAT and CDDD are covered by the SMILES family already present.

Two caveats for these molecules. The corpora are drug-like, 20 to 40 heavy atoms. Ethanol and isobutanol have 3 and 5, so ECFP is nearly empty and a transformer embedding is poorly determined for them, which is part of why isobutanol has no chemical neighbor anywhere (Section 6). The salts (sodium acetate, the imidazolium chlorides) are multi-fragment SMILES that fingerprints handle and Uni-Mol cannot conformer.

Reaction-aware encoders were left out on purpose. A reaction-hypergraph embedding places a metabolite by the reactions it takes part in, which is the right context for a Yeast9 metabolite and no context at all for an inhibitor or a drug whose fate in the cell is not in the model. When the medium components enter the environment encoding (Section 7) the metabolites among them can carry a reaction branch; the inhibitors cannot.

Coverage. Every one of the 343 compounds has a SMILES in the pinned identity table, and eleven of the twelve encoders embed all of them (Table 7). Uni-Mol embeds 332: its wrapper library substitutes flat or all-zero coordinates when a conformer cannot be generated (metal dihalides, isolated ions, nitric oxide, and bleomycin at 180 atoms) and would have embedded eleven Hillenmeyer compounds from nothing. The encoder’s `check` method now reruns the conformer step and raises on either signature, so those eleven are counted as failures rather than served as vectors. None of them is a Vanacloig compound. One dosed Hillenmeyer name, tunicamycin, is a homologue mixture with no InChIKey and is counted as not a compound.

Cost. The fingerprints, Mol2Vec and MolE embed the union in well under a second; the SMILES transformers in under a second; the RDKit descriptors in two seconds; Uni-Mol in fifty because of conformer generation. At the scale of these datasets no encoder is expensive, and the choice can be made on what the vectors do.

Table 7: Coverage of every encoder over the compounds of the three datasets (distinct InChIKeys with a dosed record: 41, 114 and 290; 343 in the union). Seconds are wall time over the union on one shared GPU, model load excluded.

encoder	dim	Vanacloig 2022	Hillenmeyer HOM	Hillenmeyer HET	s
chemberta2_mlm	384	41 of 41	114 of 114	290 of 290	0.4
chemberta2_mtr	384	41 of 41	114 of 114	290 of 290	0.2
ecfp4_bit	2,048	41 of 41	114 of 114	290 of 290	0.1
ecfp4_count	2,048	41 of 41	114 of 114	290 of 290	0.1
fcfp4_count	2,048	41 of 41	114 of 114	290 of 290	0.1
maccs	167	41 of 41	114 of 114	290 of 290	0.3
mol2vec	300	41 of 41	114 of 114	290 of 290	0.1
mole_static	1,000	41 of 41	114 of 114	290 of 290	0.3
molformer_xl	768	41 of 41	114 of 114	290 of 290	0.7
rdkit_2d	217	41 of 41	114 of 114	290 of 290	2.7
roberta_zinc_480m	768	41 of 41	114 of 114	290 of 290	0.8
unimol_v1	512	41 of 41	103 of 114	283 of 290	65.4

6 Chemical similarity across the datasets $\times$

Exact overlap is two compounds per arm (Section 2). The softer question is whether Vanacloig’s inhibitors have chemical neighbors in Hillenmeyer, and whether chemical similarity carries any information about response similarity, because that is the route by which a model could use Hillenmeyer on inhibitors it never dosed (`chemical\_similarity.py`). Both are measured for every encoder: Tanimoto similarity for the fingerprints, cosine similarity for the dense embeddings.

Nearest neighbors. Under count ECFP4 the median Vanacloig compound’s nearest HOM compound is at Tanimoto 0.33, and only three Vanacloig compounds have a HOM neighbor above 0.5: the two exact matches and cinnamic acid against benzaldehyde at 0.54. Isobutanol’s nearest HOM compound is sorbitol at 0.14, which is no neighbor at all. HET is richer for the phenolics because it dosed ferulic acid: ferulamide (0.77), caffeic acid (0.66), vanillin (0.62), vanillic acid (0.59), sinapic acid (0.56), acetovanillone (0.52) and p-coumaric acid (0.51) all have ferulic acid as their nearest HET compound, and 4-hydroxybenzoic acid has 4-aminobenzoic acid at 0.64. The cosine nearest-neighbor similarities of the dense encoders (0.8 to 0.95) are not comparable to Tanimoto: cosine between mean-pooled transformer states is high for any two molecules, so only the fingerprint numbers say how close the chemistry is.

Table 8: Per encoder and Hillenmeyer arm: the median similarity of each Vanacloig compound to its nearest partner compound, the number of Vanacloig compounds with a neighbor above 0.5 (meaningful for the Tanimoto rows only), the Spearman between chemical similarity and cross-dataset response Spearman over every condition pair, and the median response Spearman of the top decile of pairs by chemistry against the rest.

encoder	partner	NN sim. median	V with NN > 0.5	chem vs response ρ	p	top-decile ρ	rest ρ
chemberta2_mlm	hom	0.85	41	0.064	1.3e-05	0.019	0.011
chemberta2_mtr	hom	0.93	41	0.054	2.4e-04	0.015	0.011
ecfp4_bit	hom	0.33	7	0.089	1.3e-09	0.011	0.012
ecfp4_count	hom	0.33	3	0.080	4.7e-08	0.014	0.011
fcfp4_count	hom	0.39	8	0.038	9.8e-03	0.010	0.012
maccs	hom	0.55	30	0.041	5.4e-03	0.014	0.011
mol2vec	hom	0.95	41	0.078	8.8e-08	0.020	0.011
mole_static	hom	0.91	41	0.008	5.7e-01	0.011	0.012
molformer_xl	hom	0.81	41	0.079	7.0e-08	0.022	0.011
rdkit_2d	hom	0.71	38	0.016	2.7e-01	0.012	0.011
roberta_zinc_480m	hom	0.83	41	-0.028	5.6e-02	0.009	0.012
unimol_v1	hom	0.90	41	0.042	6.4e-03	0.012	0.011
chemberta2_mlm	het	0.85	41	0.037	4.8e-05	0.005	0.002
chemberta2_mtr	het	0.94	41	0.031	6.7e-04	0.005	0.002
ecfp4_bit	het	0.33	11	0.014	1.4e-01	0.003	0.002
ecfp4_count	het	0.32	11	0.033	3.3e-04	0.004	0.002
fcfp4_count	het	0.44	15	0.039	2.6e-05	0.005	0.002
maccs	het	0.60	33	0.002	8.6e-01	0.003	0.002
mol2vec	het	0.94	41	0.015	9.8e-02	0.003	0.002
mole_static	het	0.93	41	-0.005	5.6e-01	0.002	0.002
molformer_xl	het	0.82	40	0.040	1.1e-05	0.006	0.002
rdkit_2d	het	0.72	39	0.036	7.2e-05	0.005	0.002
roberta_zinc_480m	het	0.85	41	0.011	2.1e-01	0.003	0.002
unimol_v1	het	0.90	41	0.033	3.4e-04	0.004	0.002

Chemistry against response. Over the 4,674 Vanacloig-by-HOM condition pairs, chemical similarity and response similarity correlate at Spearman 0.08 to 0.09 for count and bit ECFP4, MolFormer-XL and Mol2Vec ($p < 10^{-7}$), at

0.05 to 0.06 for the ChemBERTa-2 heads, and not at all for MolE, the RDKit descriptors and the ZINC RoBERTa (Table 8, Figure 6). The effect is real and it is small: the top decile of pairs by chemistry has a median response Spearman of 0.011 to 0.022 against 0.011 for the rest. Against HET every correlation is below 0.04. Figure 6 shows why the number is small: under ECFP4 almost every pair sits at a Tanimoto below 0.25, where the response similarity is centered on zero with a spread of ± 0.1 , and the few pairs above 0.5 do not stand out from it.

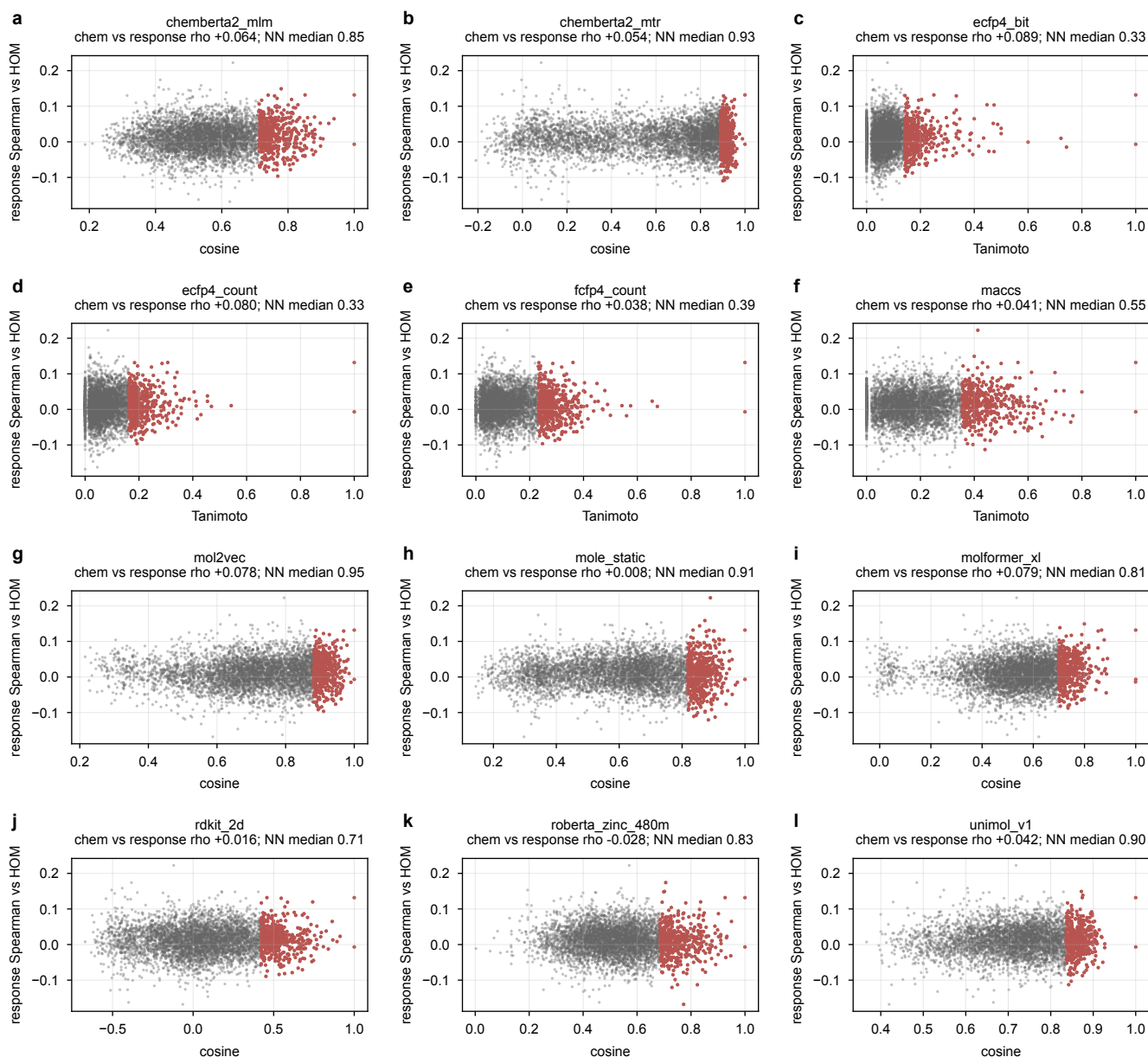


Figure 6: Chemical similarity against cross-dataset response Spearman, one point per (Vanacloig condition, HOM condition) pair, for every encoder; the top decile of pairs by chemistry in red. Panels (a) to (l) are the twelve encoders in alphabetical order; the title carries the Spearman between the two axes and the median nearest-neighbor similarity.

What this says for transfer. A model that represents compounds with ECFP4 or a SMILES transformer gets a weak prior from Hillenmeyer about which genes an unseen inhibitor will hurt, weak because chemically close pairs are barely more response-similar than distant ones. That is the expected outcome for a dataset whose compounds are drugs and metabolic poisons chosen for coverage of mechanism, against a panel of small hydrolysate molecules: the two compound sets occupy different regions of chemical space, and the pairs that would test the prior at high similarity do not exist. For the phenolic inhibitors the HET arm offers ferulic acid as a neighbor at Tanimoto 0.5 to 0.77, and ferulic acid itself did not transfer at the doses used (Section 4). For isobutanol there is no neighbor in either arm. The transfer test of Section 7 therefore has a prior of a small positive effect for the fingerprint and transformer encoders and no effect for MolE, and it should be read against that prior rather than against the hope that chemistry alone bridges the two datasets.

7 What follows x

Baselines before models. The ceilings of Section 3 become scores through the same ladder the trigenic work used: a per-gene additive ridge and a nearest-neighbor model over genes, fit on Vanacloig alone and then on Vanacloig with HOM, under a split that holds out whole compounds. A held-out compound is the case a strain-design model faces, a stress it has never seen, and it is the split that isolates the model from the dataset offsets of Section 2. Every per-compound score is reported beside that compound’s ceiling, because a score of 0.3 on a compound with a ceiling of 0.9 and a score of 0.3 on one with a ceiling of 0.35 are different results.

Two transfer tests. Section 4 found one exact match that transfers. The model question is whether a partner dataset helps on held-out inhibitors through exact matches only, or through chemical similarity as well. Both are testable on the same split: add HOM restricted to the exact-match compounds, then add all of HOM with compounds encoded by one of the encoders of Section 5, and compare the held-out Vanacloig score against Vanacloig alone. Section 6 gives the prior for the second arm.

The environment encoding. The first environment representation is the dosed compound alone, one vector per record from a chosen encoder. The second adds the medium: SynBase, YPD and SC dropouts enter as sets of compound vectors rather than as names, which is the representation the metabolism arm will share. Whether adding the metabolic model degrades the held-out compound score is the ablation that decides whether the integrated representation costs anything.

Lian 2019 as the external furfural check. Lian et al. 2019 (10.1038/s41467-019-13621-4) selected a genome-wide CRISPRa, CRISPRi and CRISPRd library on furfural. Round 1 of the CRISPRd arm, about 21,000 guides over about 5,200 genes in the unmodified host, is the nearest thing to a deletion screen; the activation and interference arms have no deletion analog and rounds 2 and 3 carry integrated backgrounds. Furfural’s ceiling in Vanacloig is 0.88, so there is signal to transfer, and the environments differ (aerobic SED-URA with G418 at 5 mM against anaerobic SynBase at IC30), so a weak score would not separate the model from the modality.